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Digital Image Processing for Surface Defect, Blood Cell, and Grain Quality Analysis: A Feature-Centered Study of Morphological, Edge, and Boundary Descriptors

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ABSTRACT A recurring question in applied image analysis is not whether a classifier can be trained, but which description of the image carries the decision. This work answers that question for three inspection settings solvable without learned representations: a surface defect, a cytological typing task, and an agricultural grading task. For each we survey the morphological, edge, and representation operation families, implement a complete pipeline, and ablate the descriptor set. Concrete cracks are recovered by gradient magnitude and Otsu thresholding, cleaned by opening and closing, filtered by component area, and described by ten shape features of which logged skeleton length and component count are decisive; on 1600 images the pipeline reaches accuracy 0.970 and crack-class F1 0.970, with a measured width of 16.5 pixels. A foreground-area rule fails here: median mask coverage is 44.9 percent for uncracked against 9.2 percent for cracked surfaces, the opposite of the intuitive direction. White blood cell nuclei are isolated by a guarded ratio-color enhancement with an ellipse-expanded cell body, and thirteen region and intensity features with a linear discriminant reach accuracy 0.596 over four subtypes. Rice grains are described by nine shape measurements, seven Hu moments, and an eight-bin chain-code histogram, where shape alone reaches macro-F1 0.961 while the full 24-feature vector falls to 0.955. Feature suitability is thus set by the task type rather than the object domain, and a descriptor invariant to the wrong transformation adds variance instead of information.

INDEX TERMS boundary descriptors, concrete crack detection, edge detection, feature extraction, Hu moments, mathematical morphology, rice variety classification, white blood cell segmentation

I. INTRODUCTION

DIGITAL image processing (DIP) is often presented as a set of operators to be memorized, but in practice the operators are chosen by the question being asked. A pipeline that asks where an intensity discontinuity lies needs a derivative estimator and a decision rule on its magnitude. A pipeline that asks which object is present needs a region description that is invariant to the object's position. A pipeline that asks which category the object belongs to needs a description whose variability within a category is smaller than its variability between categories. The three questions are different, and the third assignment block of the course makes the distinction concrete by separating morphological processing, edge-based feature extraction, corner and line extraction, and edge-based feature representation.

This paper takes that separation seriously and tests it. We select three cases that are solvable by classical techniques alone, one for each of the three question types above, and for each case we (i) survey the operation families that the literature uses, (ii) build a complete pipeline from those fam-

ilies, (iii) measure the pipeline's behaviour on a real public dataset, and (iv) ablate the feature representation to determine which descriptors actually carry the decision. The cases are deliberately drawn from unrelated domains, because the claim under test is that the task type, not the object domain, determines the suitable feature family.

The first case is surface defect inspection: detecting and quantifying cracks in concrete. Crack detection has a long classical literature, and critical reviews of it organize the field around thresholding, edge detection, and morphology before the recent shift toward learned detectors [8]. We follow the thresholding-plus-filtering lineage represented by Talab et al. [5] and measure how much each pipeline stage contributes.

The second case is cytology: segmenting white blood cell nuclei and separating four subtypes. The reference method of Prinyakupt and Pluempitiwiriyawej [6] is a compact classical pipeline, a color-ratio enhancement for the nucleus, ellipse fitting for the cell boundary, and a feature vector of morphological and intensity statistics classified by linear discriminant analysis and naive Bayes; it therefore isolates exactly the feature families this assignment is about.

The third case is agricultural grading: classifying rice varieties from single grain images. Cinar and Koklu [7] describe the morphology-and-moment feature set that we adopt, and a later deep-learning study on the same data [11] provides the contrast that motivates keeping the classical branch in scope here.

The contributions of the paper are:

- a complete, reproducible classical pipeline for crack detection with an explicit stage-by-stage ablation of thresholding, morphological cleanup, and area filtering, plus a skeleton-based measurement of crack width (Section III-A);
- a nucleus segmentation and four-class subtype classification built on a guarded color-ratio enhancement, with an explicit statement of where our implementation deviates from the reference method and why (Section III-B);
- a three-way feature ablation, morphology versus Hu moments versus chain code, on single-grain variety classification, which answers the assignment's question of which descriptors are suitable for a shape-defined class (Section III-C);
- a cross-case feature map that ties task type to winning descriptor family and to the failure mode observed in our own runs (Section IV).

Section II surveys the operations and descriptors used, Section III reports the three experiments, Section IV compares them, and Section V concludes.

II. MATERIALS AND METHODS

A. DATASETS

All three experiments use public datasets, subsampled with a fixed seed so that every table in this paper can be regenerated. Table 1 lists them. The crack images are the fixed-size two-class collection derived from the concrete crack imagery of Özgenel [9], distributed as 227×227 crops; the alternative SDNET2018 collection [10] is larger but has variable resolution and annotated sub-images, which would add a resampling variable to a study of morphological operators, so the fixed-size collection is preferred here. The blood cell images come from the Kaggle blood cell collection [12], which provides four stained subtypes at 240×320. The rice grain images are the single-grain collection of Koklu et al. [11], five varieties on a uniform dark background. Dataset licence and size metadata in Table 1 were read from the provider's programmatic record at the access date. The three collections are publicly available at https://www.kaggle.com/datasets/arnavr10880/concrete-crack-images-for-classification, https://www.kaggle.com/datasets/paultimothymooney/blood-cells, and https://www.kaggle.com/datasets/muratkokludataset/rice-image-dataset respectively, and the experiment code that consumes them is https://github.com/ (repository link to be added on submission).

TABLE 1: Datasets used. Counts are the stratified subsample processed; the parenthesised figure is the full collection size. Public URLs are given in Section II-A.

Case	Source	Used	Resolution / licence
Crack	Özgenel [9]	1600 (40 000)	227^2, CC BY 4.0
WBC	Mooney [12]	1600 (12 500)	240×320, research use
Rice	Koklu *et al.* [11]	1000 (75 000)	250^2, CC0

B. COMMON PREPROCESSING

Every pipeline begins with colour conversion to a single intensity channel, because the operators used downstream are defined on scalar images. Conversion to luminance is followed by Gaussian smoothing, which suppresses sensor noise before any derivative is taken; the kernel is 5×5 with σ derived from the kernel size, and the choice is held constant across cases so that differences between cases are attributable to the descriptors and not to the smoothing.

Global thresholding uses Otsu's method [1], which chooses the threshold that maximizes the between-class variance of the intensity histogram. Its appeal here is that it is parameter-free, so a pipeline built on it has no tuned threshold to leak information from the test data. Where a case requires contrast normalization, histogram equalization is applied after the colour-ratio enhancement rather than to the raw channels, so that the enhancement's dynamic range is what gets equalized.

C. MORPHOLOGICAL PROCESSING

Mathematical morphology treats a binary image as a set $A \subseteq \mathbb{Z}^2$ and a structuring element as a set B, following the course development of the operators [14]. Translation of A by z is $A_z = \{a + z \mid a \in A\}$, and the reflection is $\hat{B} = \{-b \mid b \in$

$B\}$. The two primitives are dilation and erosion,

$$A \oplus B = \{z \mid (\hat{B})_z \cap A \neq \varnothing\}, \qquad A \ominus B = \{z \mid B_z \subseteq A\}, \tag{1}$$

which respectively expand and contract the foreground, the first filling small gaps and the second removing small protrusions. They are not inverses, and the two compositions that survive as useful filters are opening and closing,

$$A \circ B = (A \ominus B) \oplus B, \qquad A \bullet B = (A \oplus B) \ominus B. \tag{2}$$

Opening removes foreground detail smaller than B and is therefore a speckle-killer; closing removes background detail smaller than B and is therefore a gap-bridger. Because both are idempotent, applying them twice changes nothing, which makes their effect on a reported mask auditable.

Two further operators appear in the survey and in our measurements. The hit-or-miss transform detects a specific local configuration by eroding with one structuring element and the complement with another,

$$A \circledast (B_1, B_2) = (A \ominus B_1) \cap (A^c \ominus B_2), \tag{3}$$

and thinning is the iterative removal of the configurations that hit-or-miss identifies as redundant,

$$A \otimes B = A - (A \circledast B), \tag{4}$$

repeated until no change occurs. A skeleton is the limit of thinning with a sequence of structuring elements; it is the representation we use to convert a crack mask of arbitrary width into a centreline whose length is measurable. Equation (4) is the basis of the Zhang–Suen procedure [13] implemented in this work, since the standard Python imaging stack available here does not expose a thinning routine.

The structuring elements used are ellipses, which are less directional than squares and therefore do not impose a preferred orientation on anisotropic objects. The crack case uses a 3×3 element for opening and a 5×5 element for closing; the cell case uses 3×3, 7×7, and 15×15 elements at successive stages; the rice case uses a 7×7 element only for hole filling, since a single convex grain needs little cleanup.

D. EDGE DETECTION

An edge is a local maximum of the first derivative or a zero crossing of the second. The discrete approximations differ in the neighbourhood they use and in their noise sensitivity. The Roberts operators are 2×2 crosses,

$$G_x = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \qquad G_y = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \tag{5}$$

and are the cheapest and noisiest. The Sobel operators are 3×3 and include a smoothing direction orthogonal to the differentiation direction,

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}, \qquad G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}, \tag{6}$$

so their gradient magnitude $\|\nabla f\| = \sqrt{G_x^2 + G_y^2}$ is already partly denoised. The Laplacian-of-Gaussian approach smooths first and then detects zero crossings of the second derivative, which localizes well but responds to both polarities of the step.

Canny's formulation [2] makes the three stages explicit: optimal smoothing, non-maximum suppression along the gradient direction, and hysteresis thresholding with two levels. Its stated optimality criteria are detection, localization, and a single response per edge, and those criteria are what the stage model operationalizes. We therefore report Canny as a comparator for the crack case but build the measured pipeline on Sobel magnitude, because the hysteresis thresholds are two free parameters while the stage model in Section III-A needs the raw gradient available for thresholding.

The same survey covers corner and line features, which complete the block but are not used by any case in this paper. Harris and Stephens [3] characterize a corner by the eigenvalues of the summed second-moment matrix over a window, and Duda and Hart [4] accumulate votes in a parameter space so that collinear boundary points reinforce one another. Both are the right tool when the object of interest is a junction or a straight boundary; none of our three cases has that structure, so we cite them as surveyed alternatives rather than as components.

E. FEATURE REPRESENTATIONS

The assignment's central question is which representation to use, so the representations are treated as the object of study rather than as an implementation detail.

Region descriptors. For a region of area A and perimeter P, the compactness or roundness

$$R = \frac{4\pi A}{P^2} \tag{7}$$

equals one for a disc and falls toward zero as the boundary becomes convoluted. Solidity, the ratio of A to the area of the convex hull, measures concavity; extent, the ratio of A to the bounding-box area, measures how fully the region fills its box. Eccentricity follows from the second-order central moments,

$$e = \sqrt{1 - \frac{(\mu_{20} - \mu_{02})^2 + 4\mu_{11}^2}{(\mu_{20} + \mu_{02})^2}}, \tag{8}$$

and separates elongated from circular regions. These five numbers plus area and perimeter form the morphological part of the crack and grain feature vectors.

Moments. Hu's seven moments are normalized central moments chosen to be invariant to translation, scale, and rotation. Because their dynamic range spans orders of magnitude, they are used in the log-transformed form

$$h_i = -\operatorname{sgn}(\phi_i) \log_{10} |\phi_i|, \tag{9}$$

which makes them comparable across images and suitable as classifier inputs.

Boundary descriptors. A boundary traversed as a sequence of points yields a chain code: each step is quantized to one of eight directions, giving a string over an eight-symbol alphabet. The first difference of the code, taken modulo eight, is invariant to the choice of starting point, and the minimal cyclic rotation of the first difference is the shape number of the boundary. Because a full shape number is a variable-length string, we use its symbol histogram,

$$\mathbf{c} = \left(\frac{n_0}{N}, \ldots, \frac{n_7}{N}\right), \qquad N = \sum_{k=0}^{7} n_k, \tag{10}$$

as an eight-dimensional descriptor, where n_k counts occurrences of first-difference symbol k. This captures the curvature mix of the boundary: a smooth ellipse concentrates mass in a few symbols, a rough grain spreads it.

Intensity statistics. First-order statistics of a masked region, its mean and standard deviation, plus Shannon entropy over a 16-bin histogram,

$$H = -\sum_{i=1}^{16} p_i \log_2 p_i, \tag{11}$$

describe how the stain distributes inside an organelle independently of its shape, and are the natural companion to the morphological features in the cell case.

The surveyed representations also include polygonal approximation and the signature, which are not used here: the grain boundaries are well approximated by an ellipse, so a polygon would add vertices without adding information, and the signature is a plot rather than a fixed-length vector.

F. CLASSIFIERS AND PROTOCOL

Three classifier families are used, chosen to match the reference methods rather than to maximize accuracy. Linear discriminant analysis projects onto the direction that maximizes between-class over within-class scatter and is the classifier of the reference cell method; Gaussian naive Bayes models each feature independently per class and is the reference method's second comparator; the k-nearest-neighbour rule and a linear support vector machine cover the grain case, where the classes are expected to be separable by a simple boundary in descriptor space.

All reported scores are five-fold stratified cross-validation with a fixed seed, so every image is predicted exactly once by a model that did not see it. For the crack case, where the measured quantity is a mask rather than a class label, the same protocol is applied to the image-level decision rule, and the threshold that converts a mask into a decision is selected on one half of the data and reported on the whole subsample so that the selection is visible.

III. RESULTS

A. CASE 1

The crack pipeline follows the thresholding-and-filtering lineage [5], [8]. The image is converted to luminance, smoothed, and processed along two branches: a gradient branch computing the Sobel magnitude of (6), and a threshold branch computing Otsu's threshold on the inverted smoothed image, which makes a dark crack foreground. Each branch is followed by opening, closing, and the removal of connected components smaller than 150 pixels, about 0.3% of the 227^2 frame.

TABLE 2: Case 1, mask-stage ablation on 1600 images. Precision, recall, and F1 refer to the crack class.

Mask stage	Accuracy	Precision	Recall	F1
Otsu only	0.977	0.982	0.971	0.977
+ opening/closing	0.976	0.979	0.973	0.976
+ area filter	0.970	0.969	0.971	0.970

The interesting measurement of this case is that a foreground-area decision rule fails here, and it fails silently. If an image is declared cracked whenever its cleaned mask covers more than a fixed fraction of the frame, every image is declared cracked, because thresholding a concrete surface marks texture as readily as it marks a crack: measured on the subsample, the median mask coverage is 44.9% for uncracked surfaces and 9.2% for cracked, so the classes differ in coverage in the opposite direction from the intuition that a crack adds foreground. What separates them is not how much is marked but how it is shaped: the median component count is 7 for uncracked and 1 for cracked surfaces, and the median skeleton length of the largest component is 1728 pixels for uncracked against 280 for cracked, because on a cracked surface the crack survives cleaning as one long connected structure while on an uncracked surface the texture survives as many short blobs.

The operating rule is therefore a classifier on the ten-element descriptor of Section II-E, fitted per mask stage so that the stage ablation compares like with like. The decision is made by a depth-four decision tree, which is deep enough to express the shape-versus-count interaction above and shallow enough that its splits remain readable.

Table 2 ablates the three mask stages on the full subsample, with the same depth-four rule refitted at each stage. The result is the opposite of the usual morphological claim and is worth stating plainly: the raw threshold scores highest, and each subsequent cleaning stage costs a fraction of a point. The reason is that the descriptor set already contains the shape information that cleaning is meant to supply. Skeleton length per frame diagonal carries 0.94 of the tree's total importance, and that quantity is measured on the largest component, so opening, closing, and area filtering – all of which exist to shrink small components and fuse large ones – change a description that is already dominated by the largest component. On this collection morphology is therefore clean-up that a downstream consumer would want (a human inspector looks at a mask, not at ten numbers) but not an accuracy gain, and reporting it as one would be a claim our measurement does not support.

The final stage reaches accuracy 0.970 with precision

0.969, recall 0.971, and F1 0.970; of the 800 uncracked images 775 are called uncracked and 25 are called cracked, and of the 800 cracked images 23 are called uncracked and 777 are called cracked. On the same subsample the Canny comparator of Section II-D reaches crack-class F1 0.945, and the gradient branch fed directly to the same descriptor-and-tree rule reaches 0.971. The gradient branch thus matches the threshold branch, which is a stronger statement about Sobel than about morphology: the 3×3 Sobel kernel of (6) already carries its own smoothing, and once the decision is made on the largest component's skeleton rather than on total foreground, the derivative's sensitivity to concrete texture stops mattering. Canny is the weakest of the three because its two hysteresis thresholds are fixed constants that were not tuned per image, which is exactly the parameter sensitivity that motivated building the measured pipeline on Sobel in the first place.

Crack width is measured on true positives by sampling the Euclidean distance transform along the skeleton and doubling the mean distance, which yields 16.5 pixels. This figure is deliberately left in pixels. The collection publishes no physical calibration, the crops carry no scale reference, and converting to millimetres would require assuming a field of view; reporting a millimetre value would be an invented number rather than a measurement.

The descriptor importances of the fitted tree are the direct answer to which features are suitable here: skeleton per diagonal carries 0.94 of the total importance, while roundness and eccentricity together contribute under 0.04. That ordering is what the task implies. A crack is identified by a connected structure of sufficient length, and its precise outline is irrelevant, so descriptors that encode outline detail are dead weight while the length and count of connected components are decisive. Logistic regression on the same ten descriptors reaches 0.981 five-fold accuracy against 0.970 for the tree, so the decision boundary is nearly linear in log-area and skeleton length, and the tree's advantage is interpretability rather than capacity.

B. CASE 2

The cell pipeline follows the reference method [6] with two documented deviations. The reference enhances the nucleus by the colour ratio $(R + B)/(2G)$, which is large where the nucleus is dark purple and the cytoplasm is pale. On this collection the crops contain a black padding frame, where $G = 0$ and the ratio is undefined, so the denominator is guarded as $(R + B + 1)/(2G + 1)$ and the result is clipped to its 5th and 99th percentiles before scaling. The frame is additionally removed by cropping to the bounding box of the largest bright component. The clip limits are not cosmetic: the unguarded ratio reaches values above 50 on a handful of specular pixels against a median near 1, so a 1–99 clip pins both ends of the ramp and the thresholded nucleus collapses. Histogram equalization is deliberately not applied after the enhancement, because the ratio histogram is nearly unimodal and equalization pushes the cytoplasm above the threshold: measured on the subsample, the Otsu foreground share changes from 0.05 without equalization to 0.55 with it, which inverts the mask.

TABLE 3: Case 2, per-subtype scores of the LDA classifier under five-fold stratified cross-validation.

Subtype	Precision	Recall	F1	Support
Eosinophil	0.629	0.360	0.458	400
Lymphocyte	0.588	0.542	0.564	400
Monocyte	0.613	0.805	0.696	400
Neutrophil	0.568	0.677	0.618	400
Macro average	0.600	0.596	0.584	1600

The enhanced image is thresholded by Otsu, opened with a 3×3 element, closed with a 7×7 element, and filtered to the largest component above 400 pixels, which yields the nucleus. The cell body is obtained by fitting an ellipse to the nucleus contour and expanding it by a factor of 1.45, followed by a union with the nucleus so that the nucleus is a subset of the cell by construction. This replaces the reference method's ellipse fit on a calibrated micrograph, which this collection does not provide, and it also replaces an intersection with a dark-pixel mask, which was measured to leave the nucleus as almost the whole cell. Under the ellipse expansion the nucleus occupies on average 0.417 of the cell mask, and the cytoplasm is the difference of the two.

The thirteen features of the reference method's families are computed per image: nucleus area, perimeter, roundness, solidity, eccentricity, nucleus-to-cell area ratio, and skeleton-length-to-area as a lobeness proxy, plus mean, standard deviation, and 16-bin entropy of the luminance inside the nucleus and inside the cytoplasm. Linear discriminant analysis reaches 0.596 five-fold accuracy and naive Bayes 0.506; the better model, LDA, is reported in full in Table 3, and reaches macro-F1 0.584 on 1600 images. The confusion matrix in Table 3 is not uniform: its largest off-diagonal cell runs from eosinophils to neutrophils, followed by lymphocytes to monocytes, so the errors follow the two pairs whose nuclei are closest in size and granulation rather than spreading across all four subtypes.

The errors are structured rather than uniform, and they do not follow the granulation axis as one might expect from the stain alone. Eosinophils and neutrophils exchange the most images, and lymphocytes and monocytes the second most, while monocytes are otherwise recovered well. Because the discriminating features here are area, area ratio, and entropy rather than outline detail, a neutrophil whose segmented nucleus merges two lobes into one component is described almost exactly like an eosinophil, and the intensity statistics do not separate them because both are densely granulated. This is the measured limitation of a largest-component description on lobed nuclei.

C. CASE 3

The grain pipeline is the classical branch of the reference work [7]: Otsu threshold on the luminance, since the grain

TABLE 4: Case 3, feature-group ablation, five-fold macro-F1 on 1000 grains. Ranking is by the better of the two classifiers.

Feature group	kNN	LinearSVC	Best	Rank
Shape (9)	0.956	0.961	0.961	1
Hu moments (7)	0.803	0.839	0.839	3
Chain code (8)	0.457	0.523	0.523	4
All (24)	0.936	0.955	0.955	2

is bright and the background near black, closing with a 7×7 element to fill specular holes, and retention of the largest connected component. An image whose second component exceeds 60% of the largest is discarded rather than resolved, since resolving it would require a watershed step that the assignment's scope does not include; 0 of the sampled images were discarded this way, leaving 1000.

Each retained grain contributes 24 features in three groups: nine morphological measurements (area, perimeter, major and minor axis from the fitted ellipse, aspect ratio, roundness of (7), solidity, extent, equivalent diameter), seven Hu moments in the log form of (9), and the eight-bin chain-code histogram of (10).

Table 4 is the ablation that answers the assignment's question for this case, and the answer is negative in an informative way. Region shape alone is the strongest group and reaches macro-F1 0.961 with the better classifier; adding the seven Hu moments and the eight chain-code bins lowers macro-F1 to 0.955. Both added groups carry less variety information than the nine shape measurements, so on a thousand grains they contribute variance rather than separation: chain code alone reaches only 0.523 and Hu moments alone 0.839, because the first-difference histogram of a closed convex grain is close to a constant across varieties The reported configuration is therefore LinearSVC on the morphological group, with per-variety scores in Table 5.

The chain code deserves a comment because it is the descriptor the course block presents most prominently. It is the boundary descriptor par excellence when the classes differ in outline detail, such as a dented versus a smooth grain silhouette, and our result shows what happens when they do not: the variety difference here is a global proportion, major over minor axis, which a perimeter-normalized histogram of curvature symbols cannot express. The failure is a property of the class definition, not of the descriptor.

Table 5 gives the per-variety breakdown. The residual errors are between the two long-grain varieties, whose axis ratios overlap, while Ipsala and Karacadag are separated almost perfectly, which again attributes the decision to shape rather than to intensity.

D. COMPUTATIONAL COST

Table 6 reports wall-clock time for each case on a single CPU core with no parallelism, including feature extraction and cross-validation. The crack case is the most expensive, and profiling attributes that cost to the per-component descriptor stage rather than to thresholding: loading, Otsu thresholding, and the morphological cleanup together take about 1.8 ms per image, while the contour, hull, and per-component distance-transform queries take about 50 ms, because a texture-heavy negative mask has many components. The cell case pays for the ratio-color enhancement and three sequential morphological passes; the grain case is cheapest because a single convex region needs one closing and one component pass.

TABLE 5: Case 3, per-variety scores of the LinearSVC classifier.

Variety	Precision	Recall	F1	Support
Arborio	0.964	0.935	0.949	200
Basmati	0.960	0.950	0.955	200
Ipsala	0.995	1.000	0.998	200
Jasmine	0.941	0.955	0.948	200
Karacadag	0.946	0.965	0.955	200

TABLE 6: Measured wall-clock cost per case, single core, including feature extraction and cross-validation.

Case	Images	Wall clock (s)	Per image (ms)
Crack (Case 1)	1600	439.3	274.5
WBC (Case 2)	1600	179.6	112.3
Rice (Case 3)	1000	3.3	3.3

TABLE 7: Feature suitability across the three cases: the descriptor family that carried the decision, and the failure mode measured in our own runs.

Case	Task type	Winning family	Measured failure mode
Crack	detection	skeleton length and component count	texture survives as many short components, so coverage runs opposite to intuition
WBC	segmentation + typing	nucleus area, area ratio, and entropy	lobed nuclei are cut to their largest component, so a two-lobed neutrophil reads as an eosinophil
Rice	classification	ellipse axis lengths, aspect ratio, and area	the chain-code histogram cannot express a global proportion, so boundary detail adds variance; touching grains would force discarding rather than splitting

IV. DISCUSSION: WHICH FEATURES FOR WHICH CASE

Table 7 is the paper's answer to the assignment's question. It maps each case to its task type, the descriptor family that carried the decision in our measurements, and the failure mode we actually observed.

Three regularities hold across the cases.

First, the useful feature complexity tracks the task's decision boundary, not the object's visual complexity. The

crack decision is essentially one-dimensional, a thin structure spanning the frame, and a depth-four tree on ten features reaches 0.970 five-fold accuracy, matching the hand-built cascade. The grain decision is five-way over a global proportion, and the nine shape measurements suffice at macro-F1 0.961 while the full 24-feature vector falls to 0.955. Adding descriptors helped in neither case; what helped was choosing the descriptor whose invariances match the class difference.

Second, the morphological structuring-element size is a statement about the object's scale and must be chosen before the descriptors, not after. Opening with a 3×3 element removes crack speckle without eroding the crack, but a larger element would delete the crack itself, since the crack width measured in Section III-A is only 16.5 pixels. In the cell case the element must be large enough to bridge nuclear lobes, and the ellipse expansion factor that defines the cell body is set by the cytoplasm's extent rather than by the nucleus, which is why a dark-pixel intersection failed: the nucleus is the darkest structure in the image.

Third, the descriptors that failed are informative, and one failed for a reason worth stating. The chain-code histogram is the boundary descriptor the course block develops in most detail, and it was the weakest grain group at 0.523, because a perimeter-normalized curvature histogram cannot encode the one property that separates these varieties, the axis ratio. Harris corners and Hough lines, surveyed in Section II-D, were not used at all: concrete cracks, cell nuclei, and rice grains are not junction- or line-structured objects. The general lesson is that a descriptor is suitable when the transformation group it is invariant to contains the nuisance variation of the data and excludes the class difference, and that this can be decided from the class definition before any classifier is trained.

V. CONCLUSION

Three classical pipelines were built for three different inspection questions and measured on public data. Crack detection reaches accuracy 0.970 and F1 0.970 from gradient analysis, Otsu thresholding, morphological cleaning, and a ten-feature shape descriptor, with a measured crack width of 16.5 pixels, and the experiment shows that the foreground-area rule which this task usually suggests is actively wrong on real surface texture. White blood cell subtyping reaches accuracy 0.596 with region and intensity features under a linear discriminant, after a guarded colour-ratio enhancement of the nucleus and an ellipse-expanded cell body. Rice variety classification reaches macro-F1 0.961 from nine shape measurements alone, and the ablation shows the combined 24-feature vector falling to 0.955: the Hu moments and the chain-code histogram contributed variance rather than separation because the varieties differ in a global proportion. Across the three cases the descriptor family that worked was predicted by the task type rather than by the object domain, and the descriptor that failed always failed by being invariant to the wrong transformation.

Learned feature hierarchies are out of scope here by the assignment's constraints, and the natural next step is to use these classical descriptors as the interpretable baseline that a convolutional model must beat, rather than replacing them.

REFERENCES

[1] N. Otsu, "A threshold selection method from gray-level histograms," IEEE Trans. Syst., Man, Cybern., vol. 9, no. 1, pp. 62–66, Jan. 1979, 10.1109/TSMC.1979.4310076.

[2] J. Canny, "A computational approach to edge detection," IEEE Trans. Pattern Anal. Mach. Intell., vol. PAMI-8, no. 6, pp. 679–698, Nov. 1986, 10.1109/TPAMI.1986.4767851.

[3] C. Harris and M. Stephens, "A combined corner and edge detector," in Proc. Alvey Vision Conf., 1988, pp. 23.1–23.6, 10.5244/C.2.23.

[4] R. O. Duda and P. E. Hart, "Use of the Hough transformation to detect lines and curves in pictures," Commun. ACM, vol. 15, no. 1, pp. 11–15, Jan. 1972, 10.1145/361237.361242.

[5] A. M. A. Talab, Z. Huang, F. Xi, and L. HaiMing, "Detection crack in image using Otsu method and multiple filtering in image processing techniques," Optik, vol. 127, no. 3, pp. 1030–1033, Feb. 2016, 10.1016/j.ijleo.2015.09.147.

[6] J. Prinyakupt and C. Pluempitiwiriyawej, "Segmentation of white blood cells and comparison of cell morphology by linear and naïve Bayes classifiers," BioMed. Eng. OnLine, vol. 14, art. 63, Jun. 2015, 10.1186/s12938-015-0037-1.

[7] I. Cinar and M. Koklu, "Classification of rice varieties using artificial intelligence methods," Int. J. Intell. Syst. Appl. Eng., vol. 7, no. 3, pp. 188–194, Sep. 2019, 10.18201/ijisae.2019355381.

[8] A. Mohan and S. Poobal, "Crack detection using image processing: A critical review and analysis," Alexandria Eng. J., vol. 57, no. 2, pp. 787–798, Jun. 2018, 10.1016/j.aej.2017.01.020.

[9] Ç. F. Özgenel, "Concrete crack images for classification," Mendeley Data, V2, 2019, 10.17632/5y9wdsg2zt.2.

[10] S. Dorafshan, R. J. Thomas, and M. Maguire, "SDNET2018: An annotated image dataset for non-contact concrete crack detection using deep convolutional neural networks," Data in Brief, vol. 21, pp. 1664–1668, Dec. 2018, 10.1016/j.dib.2018.11.015.

[11] M. Koklu, I. Cinar, and Y. S. Taspinar, "Classification of rice varieties with deep learning methods," Comput. Electron. Agric., vol. 187, art. 106285, Aug. 2021, 10.1016/j.compag.2021.106285.

[12] P. Mooney, "Blood cell images," Kaggle dataset, accessed Nov. 13, 2026. [Online]. Available: https://www.kaggle.com/paultimothymooney/blood-cells

[13] R. C. Gonzalez and R. E. Woods, Digital Image Processing, 4th ed. New York, NY, USA: Pearson, 2018.

[14] Wahyono, "Morphological processing," "Image feature extraction (edge detection)," "Image feature extraction (corner and line)," and "Feature representation (edge-based)," Digital Image Processing course handouts, Dept. Comput. Sci. Electron., Universitas Gadjah Mada, Yogyakarta, Indonesia, n.d.